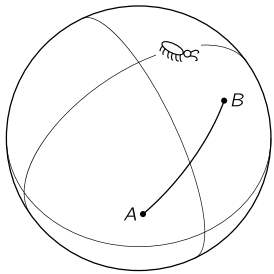

Figure 6-5 Making a "straight line" on a sphere.

the bugs have all learned how to make two lines intersect at right angles. (You can figure out how they could do it.) Then our first bug (the one on the normal plane) finds an interesting fact. If he starts at the point *A* and makes a line 100 inches long, then makes a right angle and marks off another 100 inches, then makes another right angle and goes another 100 inches, then makes a third right angle and a fourth line 100 inches long, he ends up right at the starting point as shown in Figure 6-7(a). It is a property of his world—one of the facts of his "geometry."

Then he discovers another interesting thing. If he makes a triangle—a figure with three straight lines—the sum of the angles is equal to 180°, that is, to the sum of two right angles. See Figure 6-7(b).

Then he invents the circle. What's a circle? A circle is made this way: you rush off on straight lines in many many directions from

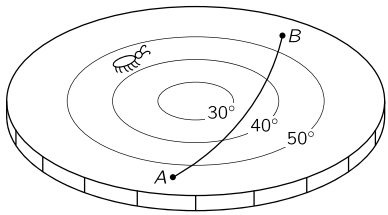

Figure 6-6 Making a "straight line" on the hot plate.